

Lokesh Sandu

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SUMMARY

System Integration and Validation Engineer with 2+ years of experience in EV off-board/on-board charging systems, powertrain, VCU integration, and HV/LV validation. Strong expertise in hardware–firmware integration, system-level testing, root cause analysis, and field issue debugging for high-voltage EV systems. Hands-on with AC/DC charging logic, CCS/IEC standards, CAN diagnostics, power electronics validation, and Python-based test analysis. Passionate about delivering safe, reliable, and scalable charging infrastructure.

PROFESSIONAL EXPERIENCE

QUCEV Technology

Feb 2024 - Present

Engineer - System Integration Engineer

- Led end-to-end hardware and firmware integration of VCU, off-board/on-board charging, and HV/LV subsystems for a 3.5-ton electric vehicle platform.
- Developed and executed system-level validation test plans for AC/DC charging enable/disable, HV pre-charge, active discharge, and contactor sequencing, including fault scenarios.
- Performed hands-on debugging of hardware and firmware issues on high-voltage systems using oscilloscopes, CAN tools (CANoe), bench setups, and electronic loads.
- Conducted failure analysis and root cause investigation for charging and HV faults observed during vehicle testing, end-of-line validation, and field operation, driving corrective actions.
- Supported hardware bring-up, signal validation, calibration, and regression testing after firmware and hardware updates using ETAS INCA & A2L.
- Validated charging safety mechanisms, including plug interlock (E-Lock) control via H-bridge PWM and diagnostic fault handling.
- Collaborated cross-functionally with charger, BMS, power electronics, firmware, and hardware teams to ensure functional, performance, and safety requirements were met.

Impact:

Improved charging robustness, reduced false charge disable faults, and accelerated issue closure during vehicle validation cycles.

CLN Energy

May 2023 - Dec 2023

GET - Embedded Firmware Engineer

- Integrated and validated FOC (Field-Oriented Control) firmware on STM32G431 (G4 series) using STM Motor Control Workbench, STM32CubeMX, and CubeIDE.
 - Supported hardware–firmware integration of Motor Control Unit (MCU), including gate drivers, current sensing, protection logic, and inverter interfaces.
 - Designed and executed unit, regression, and bench-level test cases for motor control, diagnostics, and safety-critical functions (pre-charge, active discharge, fault handling).
 - Implemented and validated CAN communication and diagnostics, including 150+ DTCs, fault classification, and real-time monitoring using BusMaster.
 - Developed derating strategies based on fault severity and thermal limits to ensure safe operation under degraded conditions.
 - Performed end-to-end debugging and validation through bench testing and dyno testing, including PID tuning, torque map optimization, and post-SOP regression testing.
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PROJECTS

3kW Motor Control Unit Development – CLN Energy

- Integrated and validated a 3kW MCU on STM32G431 for 2-wheeler applications.
- Developed and executed regression, and fault injection test cases for motor control functions (FOC, PID tuning).
- Improved system reliability and increased fault detection coverage by 25%.

3.5-Ton N-Category Electric Cargo Vehicle Development – QUCEV Technology

- Validated ISO 15118 CCS2 charging, safety functions, and drive features using HIL and lab setups.
- Supported ISO 26262 functional safety compliance.
- Designed and validated AC/DC charging state machine
- Covered grid detection, connector status, E-Lock, safety interlocks
- Tested per IEC 61851 / IEC 62196 scenarios
- Verified fault handling for voltage, frequency, and communication failures

HV System Safety & Diagnostics

- Validated HV enable conditions, isolation monitoring, and safe shutdown
- Implemented and tested charge disable strategies for multiple fault classes
- Performed bitwise CRC validation (CRC8 J1850) for safety-critical CAN signals

EDUCATION

BITS PILANI – Work Integrated Learning Programme (WILP)

M.TECH - Automotive Electronics (Ongoing)

Aug 2019 - May 2023

SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY

B.TECH - Electrical and Electronics Engineering

SKILLS

Programming & Scripting: Embedded C, CAPL, Python

Automotive Standards: ISO 26262, AIS 156, ISO 15118 (CCS2 Charging), IEC 61851, IEC 62196

Communication Protocols: CAN, SPI, I2C, UART

System Design & Modeling: Vehicle E/E Architecture , AC/DC Charging Architecture

Testing, Validation & Calibration: HIL Testing, Calibration, PID Tuning, Network Simulation, Test Case Development & Reporting

Automotive Tools: CANoe, vTESTstudio, Vector db++, ETAS INCA, VehicleSpy3, Picoscope

Version Control: Git Project Management: JIRA, Rational DOORS

Productivity Environments: MS Office (Excel, Word, PowerPoint)

Documents: DVP, DFMEA

CERTIFICATIONS

MBSE: Model-Based Systems Engineering - University at Buffalo

Introduction to Battery-management systems - University of Colorado Boulder

Introduction to ASPICE - EDUCBA

How to create a Jira SCRUM project - Coursera

Excel Essentials for Data Analytics - IBM

Six Sigma Principles - Kennesaw State University